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Genome-Wide Identification of the *BBX* Gene Family: *StBBX17* Positively Regulates Cold Tolerance in Potato

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Abstract

Potato is an important crop in the world and is rich in various nutrients. Common tetraploid potato is not tolerant of low temperatures and frost. Low-temperature stress severely affects the growth above-ground and the yield underground in potato. The *BBX* genes play an important role in the plant response to low-temperature stress. However, the molecular mechanism underlying the potato *StBBX* genes involved in cold stress response remains unclear. In the present study, 30 *StBBX* genes were identified in potato and divided into five groups. A total of 10 motifs and 10 cis-acting elements were obtained in all *BBX* proteins. All *StBBX* genes contained light responsive elements in the promoter, of which nine *StBBX* genes harbored low-temperature responsive elements. In total, 15 pairs of *StBBX* genes were identified in duplicated genomic regions. The gene expression patterns of all *StBBXs* were assessed in different tissues by transcriptome data. The qRT-PCR analysis indicated that six *StBBX* genes were significantly induced in response to cold stress. Subcellular localization suggested that the StBBX17 protein was localized in the nucleus. Compared with wild type (WT), the cold tolerance in *StBBX17* overexpression lines was dramatically increased. After cold treatment, the *StBBX17* overexpression lines displayed a less injured area of leaves and lower electrolyte leakage compared with the WT plants, demonstrating *StBBX17* positively regulated cold tolerances in potato. These results indicate that *StBBX* genes have important functions under cold stress, providing a theoretical reference for the breeding of cold-resistant potato.

Keywords: potato; cold; gene family analysis; overexpression; *StBBX17*

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1. Introduction

Low temperature is a key environmental factor that seriously limits plant growth, development, and geographical distribution. Plants have developed complex regulatory mechanisms to cope with low-temperature stress. Temperate plants with the ability of cold acclimation have the ability to acquire cold resistance through pre-exposure to low but nonfreezing temperatures [1,2]. The C-repeat-binding factors/dehydration-responsive element binding proteins (CBFs/DREB1s) signaling pathway, extensively explored in *Arabidopsis*, plays a crucial role in cold acclimation [3]. Many studies have shown that CBF proteins directly bind to the C-repeat element (CRT)/DRE cis-elements in the promoters of *COLD-RESPONSIVE* (*COR*) genes [3,4]. Several transcriptional factors (TFs) positively regulate *CBF* expression and increase plant freezing tolerance, including *inducer of CBF expression 1* (*ICE1*) [5], *calmodulin binding transcription activators* (*CAMTAs*) [6], and *BIN2* (*BRASSINOSTEROID-INSENSITIVE2*) [7]. However, some transcription factors suppress *CBF* expression and decrease cold resistance, including *MYB15*, *MYB30*, *ETHYLENE INSENSITIVE 3* [8], *phytochrome-interacting factor 3* (*PIF3*) [9], and *PIF4* [10]. Previous studies found that *CBF* genes regulated only 10–20% of *COR* genes in *Arabidopsis* [11,12], indicating the other transcription factors participate in cold stress via CBF-independent pathways.

The B-box (BBX) proteins are zinc-finger transcription factors that possess one or two B-box domains at the N termini, and some members also contain a CCT domain or valine-proline motif at the C terminus [13]. A total of 32 *BBX* genes were identified and divided into five subfamilies based on their domain structures in *Arabidopsis* [14]. Numerous studies revealed that the conserved CCT domain is involved in nuclear transport and transcriptional regulation [15]. BBX domain-contained proteins play crucial roles in regulating photomorphogenesis [16], flowering [17], anthocyanin accumulation [18], cold stress [19,20], salt and drought stress [21], and Fusarium wilt [22]. *BBX28* and *BBX29* restrains *ELONGATED HYPOCOTYL 5* (*HY5*) binds to the *BBX30* and *BBX31* promoters, forming a transcriptional feedback loop to fine-tune photomorphogenesis in *Arabidopsis* [23]. *SlBBX20* and *SlBBX21* bind to the *SlHY5* promoter, activate its transcription and promote UV-B photomorphogenesis in tomato [16]. *CmBBX8* binds to *CmFTL1* promoter and promotes flowering in chrysanthemum [17]. *PpBBX18* enhances anthocyanin biosynthesis, and *PpBBX21* reduces anthocyanin accumulation in pear [18]. Overexpression of *IbBBX24* significantly improves Fusarium wilt disease resistance and increases yields in sweet potato. *IbBBX24* activates *IbJAZ10* transcription and inhibits the transcription of *IbMYC2*, suggesting that *IbBBX24* plays an important role in regulating JA biosynthesis and signaling in sweet potato [22]. In apple, MIEL1 (MYB30-Interacting E3 Ligase1) and JAZ (JAZMONATE ZIM-DOMAIN) proteins are co-involved in the regulation of cold stress by the *BBX37-ICE1-CBF* module [19]. Arabidopsis *BBX7* and *BBX8* positively regulate freezing tolerance through *CRYPTOCHROME 2* (*CRY2*), *photomorphogenic 1* (*COP1*), and *HY5* [20]. Previous studies found that *SlBBX17* positively regulated cold tolerance in tomato [24]. BBX29 negatively regulates cold tolerance in Arabidopsis by a CBF-independent pathway [25]. Silencing of *SlBBX7*, *SlBBX9*, and *SlBBX20* reduces tomato cold tolerance [26]. *SlBBX31* positively regulates cold tolerance in tomato, and *SlHY5* can directly bind to the *SlBBX31* promoter [27]. In potato, 30 *StBBX* members have been identified and characterized, and the expression profiles of *StBBXs* were performed under diurnal cycle, etiolation and de-etiolations [28]. In our previous studies, overexpression of *StBBX14* in potato displayed less leaf damage and lower electrolyte leakage compared with the wild type under cold stress [29]. These studies have confirmed that the BBX gene family plays an important role in cold stress. However, the molecular functions of *StBBXs* involved in regulating potato response to cold stress remain unclear.

Potato (*Solanum tuberosum* L.) is one of the most important tuber crops worldwide. The common cultivated potato is not resistant to cold stress, which will affect the growth of potato shoots and the yield of potato. The potato originates in the harsh environment of the Andes mountains, where the potato germplasm resources are rich and possess many cold-resistant wild diploids [30]. The potato genome assembly sequences of the wild diploid *Solanum commersonii* with freezing tolerance were widely reported [31–33]. Many studies have reported that many cold resistance genes were identified and verified based on the *Solanum commersoni* genotype [34,35]. Overexpression of *Solanum commersonii SAD1* (*ScSAD1*) in cultivated potato increase its tolerance to freezing [34]. Overexpression of *ScGolS1* distinctly increases freezing tolerance and antioxidant activity in the potato cultivar 'Atlantic' [36]. Overexpression of *ScUGT73B4* improves glycosylated flavonoid accumulation and increases antioxidant capacity, resulting in enhanced freezing tolerance in potato [35]. The *Solanum acaule arginine decarboxylase gene ADC1* (*SaADC1*) transform in *S. tuberosum* cv. E3 increases freezing tolerance of potato in cold climates and enhances *CBF* gene expression [37]. *SaCBL1-like* (*calcineurin B-like protein*) improves potato freezing tolerance by activating the expression of *CBF*-regulon [30]. However, the molecular mechanism underlying potato response to cold stress remains largely unknown.

In this study, the evolutionary relationship, gene structure, chromosome location, and collinearity analysis of BBX family genes were investigated. The expression patterns of all identified *StBBX* genes were analyzed under cold stresses by qRT-PCR. To evaluate the *StBBX17* function of resistance to cold stress, the *StBBX17* gene was overexpressed in potato. This research would provide a foundation for further dissecting the evolutionary relationship and functional differentiation of *StBBX* family genes and exploring the molecular function of *StBBX* genes under cold stress in potato.

2. Materials and Methods

2.1. Identification and Phylogenetic Analysis of BBX Gene Family

The genome sequences were retrieved from the potato genome database (DM6.1). The conserved domain of BBX proteins (PF00643) was downloaded from the Pfam database (http://pfam.xfam.org/; accessed on 28 August 2022) to search *StBBX* candidate sequences. The HMMER 3.0 software with an E-value cut-off of 0.01 was used to obtain candidate *StBBXs* [38]. The BBX domain was verified using SMART (http://smart.embl-heidelberg.de/; accessed on 28 August 2022) and Pfam [39]. The MUSCLE (http://www.ebi.ac.uk/Tools/msa/muscle/; accessed on 28 August 2022) software was employed to construct multiple alignments of the predicted protein sequences in the potato, tomato and *Arabidopsis* [40]. The MEGA 11 v11.0.13 software was used to construct unrooted phylogenetic trees using the neighbor-joining method with 1000 bootstrap replicates [41]. The names of StBBXs were generated based on the evolutionary relationship between Arabidopsis and tomato.

2.2. Sequence Analysis of StBBXs

The exons and introns of *StBBX* gene structures were analyzed using Gene Structure Display Server (http://gsds.cbi.pku.edu.cn/; accessed on 1 September 2022). The cis-acting elements of *StBBX* promoters with 2000 bp regions upstream of the CDS were predicted using PlantCARE (http://bioinformatics.psb.ugent.be/webtools/plantcare/html/; accessed on 3 September 2022) [42].

2.3. Synteny Analysis and Chromosomal Localization of StBBXs

Multiple Collinearity Scan toolkit (MCScanX) was employed to identify *StBBX* gene sequence repetition events [43]. The intra-species paralogous pairs of the protein sequence were identified by BLASTP, and the parameters were set as follows: (1) alignment significance: E_VALUE (default: 1×10^{-5}), (2) MATCH_SCORE: final score (default: 50), (3) MATCH_SIZE: number of genes required to call a collinear block (default: 5) and the maximum gaps (default: 25) [44]. The relative positions of *StBBX* genes on potato chromosomes were located by EVOLVIEW (http://www.evolgenius.info/evolview/; accessed on 31 August 2022).

2.4. qRT-PCR Analysis

The potato variety 'Desiree' were cultured on MS medium containing 3% sucrose and 0.8% agar at pH 5.8. The 4-week-old plants were transplanted into a controlled environment chamber with a 16 h light/8 h dark cycle at a temperature of 23 °C/21 °C (light/dark). The 4-week-old plants of 'Desiree' was subjected to 2 °C for 0 h, 1 h, 2 h, 4 h, 8 h, 12 h, 24 h and 48 h. Total RNA in the leaves of 'Desiree' was isolated with different cold treatment times by MiNiBEST Universal RNA Extraction kit (TaKaRa, Beijing, China). The reverse transcription was performed with StarScript II RT Mix (GeneStar, Beijing, China). The real-time fluorescence quantification was performed on a CFX96 detection system (Bio-Rad, Hercules, CA, USA). The qRT-PCR reaction contained 1 μL of cDNA, 0.5 μL of forward and reverse primer, 10 μL of qPCR Master Mix and 8 μL of ddH_2O. *β-tubulin* was selected as internal reference gene. The relative gene expression was calculated using the Ct ($2^{-\Delta\Delta Ct}$) method. The experiment was performed with three biological replicates. All primers are listed in Supplementary Table S1.

2.5. Subcellular Localization and Plant Transformation

The coding sequence (CDS) of *StBBX17* was cloned into the PBWA (V)-GFP vector. The empty-GFP vector was used as negative control. All plasmids were transferred into *Agrobacterium tumefaciens* GV3101, and then injected into *N. benthamiana*. The DAPI marker was used to identify nuclei. The GFP fluorescence was observed by a laser confocal microscope (Nikon ECLIPSE Ti2) at excitation wavelength of 488 nm.

The CDS sequence of *StBBX17* was isolated from potato cDNA with specific primers. The PCR products and the vector pFGC1008 with the promoter of CaMV35S were digested with *Kpn I* and *Sac I*, and ligated into pFGC1008 vector through homologous recombination. The plasmid was transformed into *Agrobacterium tumefaciens* GV3101 using the electric shock method. The confirmed plasmids were transformed into the cold sensitive variety "Desiree". The potato miniature tubers were used to perform genetic transformation followed the previous methods [34]. All positively transgenic plants regenerated from tissue culture were validated by PCR amplification with genomic DNA primers. The gene expression of *StBBX17* was confirmed in the transgenic lines by qRT-PCR. The transgenic potato was planted in the greenhouse, and the wild type with the same conditions was used as a control. All the plants (30 days old) were used to perform cold treatment at −1 °C for 14 h and carry out phenotype investigation.

2.6. Frost Resistance Determination

Five-week-old potato seedlings were either exposed to cold stress at 4 °C or kept under normal conditions. Potato leaves were placed in test tubes with ice chips, and then transferred into a freezing bath with an initial temperature of 0 °C. Test tubes were cooled in increment of 0.5 °C every 30 min until −10 °C was reached. After the test tubes reached the experimental temperature, they were taken out of the freezing bath and put on ice overnight. The samples were gradually thawed and immersed in 25 mL deionized water, and then the samples were used to measure electrical conductivity (R1). The samples were autoclaved at 121 °C for 15 min and cooled for 24 h, then the total conductivity (R2) was measured. Relative electrolytic leakage was calculated as (R1/R2) × 100%, following a previously described method [45].

3. Results

3.1. Identification and Phylogenetic Analysis of StBBX Genes in Potato

A total of 30 *StBBX* candidate family members were obtained from potato genome (Supplementary Table S2) by HMMER software. To understand the evolutionary relationships and differences in gene functions among this gene family in three different species (potato, Arabidopsis and tomato), a phylogenetic tree was constructed after multiple alignment of amino acid sequences of 30 potato *BBX* (*StBBX*) genes, 32 Arabidopsis *BBX* (*AtBBX*) genes and 31 *Solanum lycopersicum BBX* (*SlBBX*) genes. These potato BBX members were named *StBBX1* to *StBBX31* based on the phylogenetic analysis results (Supplementary Table S3). The *BBX* gene family members of the three species were divided into five groups in the evolutionary tree (Figure 1). Group 1 contained five *StBBXs*, six *AtBBXs* and six *SlBBXs*. Group 2 contained 7 *StBBXs*, 9 *AtBBXs* and 7 *SlBBXs*. Group 3 comprised 3 *StBBXs*, 4 *AtBBXs* and 3 *SlBBXs*. Group 4 contained 8 *BBX* genes from each of the three species. Group 5 possessed 7 *StBBXs*, 5 *AtBBXs*, and 7 *SlBBXs*.

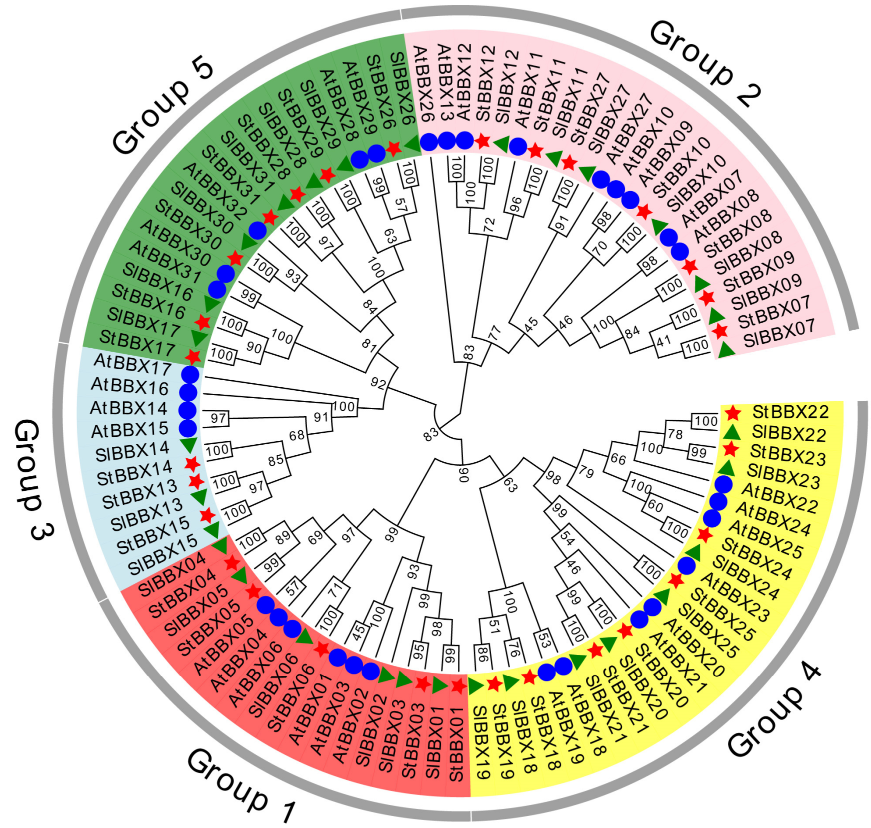

Figure 1. Phylogenetic tree of BBX gene families in potato, *Arabidopsis* and tomato. The red star represents *StBBXs*, the blue circle represents *AtBBXs*, and the green triangle represents *SlBBXs*.

3.2. *StBBX* Gene Structure and Motif Analysis

To understand the conserved structure of BBX proteins, the amino acid sequences of potato *BBX* family members were investigated by MEME website (https://meme-suite.org/meme/tools/meme, accessed on 3 September 2022) [46]. Motif analysis showed that 10 motifs were identified in the 30 *BBX* proteins (Figure 2, Supplementary Figure S1). Motif 4 and motif 2 represented the main types in the five groups. Motif 1, motif 3 and motif 7 constituted most of the B-box domains and were distributed in three different groups. Motif 5 and motif 10 only existed in Group 2 and Group 3 subgroups, respectively. Motif 6 and motif 8 were only distributed in Group 4. The B-box1 domain existed in all BBX proteins (Figure 2). The B-box2 domain mainly existed in Group 1, Group 2 and Group 4. The CCT domain was mainly distributed in Group 1, Group 2 and Group 3.

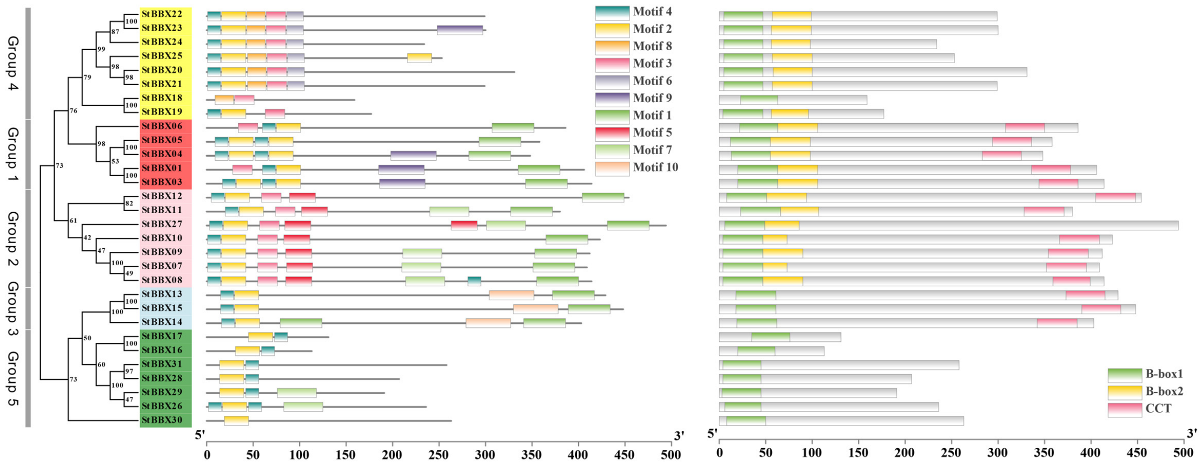

Figure 2. Evolutionary relationship, conserved motifs and protein domain analysis of StBBX gene family.

3.3. Promoter Element Prediction for StBBX Gene Family

Cis-acting element analysis detected 10 cis-acting elements in the promoter of *StBBX* genes. All *StBBX* genes contained light responsive elements in the promoter (Supplementary Table S4). Hormone regulation related to salicylic acid responsive (9 *StBBX* genes), auxin responsive (12 *StBBX* genes), MeJA responsive (14 *StBBX* genes), abscisic acid responsive (20 *StBBX* genes) and gibberellin responsive (22 *StBBX* genes) elements were identified. In all, 22 *StBBX* genes contained cis-acting regulatory elements involved in anaerobic induction, and 13 *StBBX* genes contained elements related to defense and stress responsive elements. A total of nine genes harbored low-temperature responsive and drought responsive elements, respectively, which might have played important roles in cold and drought stress.

3.4. Chromosome Location of StBBX Genes

The results for chromosomal locations showed that 30 *StBBX* genes were located on twelve chromosomes (Figure 3A). The number of *StBBX* genes in each chromosome was varied. Chromosome 12 comprised the largest number of genes, of which *StBBX05* and *StBBX20* were located on the front end, and *StBBX23*, *StBBX28*, *StBBX16* and *StBBX07* were located on the tail end. Five *StBBX* genes were distributed on chromosome 07. Four *StBBX* genes were distributed on chromosome 02 and 05. Chromosome 04 contained three *StBBX* genes. Chromosomes 01 and 06 contained two *StBBX* genes. Chromosomes 03, 08, 09 and 10 possessed only one *StBBX* gene.

3.5. Synteny Analysis of the StBBX Gene Family

To further investigate the origin and evolution of the BBX gene family, tomato and *Arabidopsis* were selected for synteny analysis by BLASTP comparison of BBX homologous sequences. In total, 52 pairs of orthologous *BBXs* containing 26 *SlBBX* and 29 *StBBX* genes between tomato and potato were identified. A total of 47 pairs of orthologous BBXs containing 23 *AtBBX* genes in *Arabidopsis* and 25 *StBBX* genes in potato were obtained (Figure 3B; Supplementary Table S5).

Gene duplication events play essential roles in generating new functional genes and promoting the evolution of species. Whole genome duplication (WGD)/segmental duplication and tandem duplication events among *StBBX* genes were assessed by MCScanX. In total, 15 WGD/segmental duplication events were identified (Supplementary Table S6, Figure S2). It was found that a single gene corresponded to multiple genes, for example, *StBBX10* corresponded to *StBBX17* and *StBBX16*.

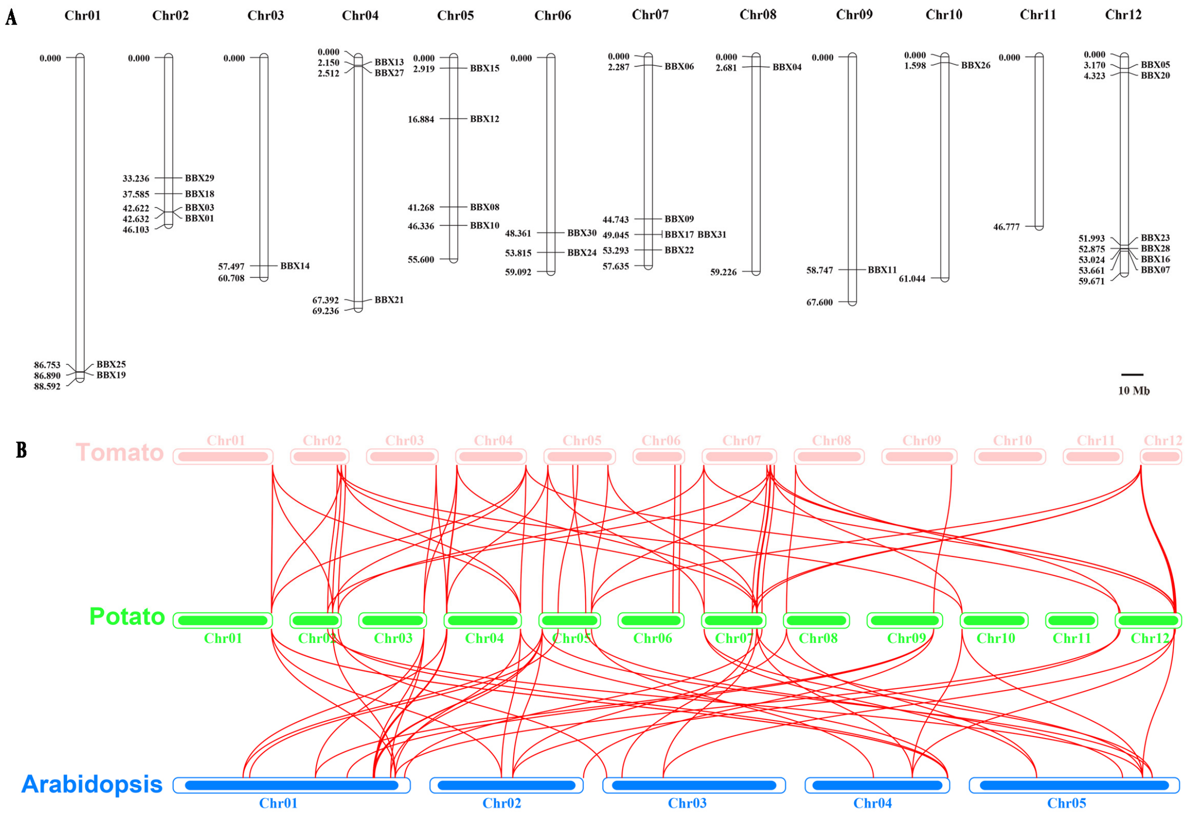

Figure 3. Chromosome location and collinearity analysis of StBBX gene family. (**A**) Chromosome location of StBBX protein. The numbers on chromosomes represent physical distances. (**B**) Orthologous collinearity of StBBX genes among tomato, potato and Arabidopsis. Collinear relationships of StBBX gene pairs are indicated with red lines.

3.6. Gene Expression Patterns of StBBXs in Different Potato Tissues

The gene expression levels of *StBBXs* in different potato tissues were evaluated by 17 RNA-seq data points in the potato genome database (cultivar, RH89-039-16). A heatmap of all *StBBX* genes with TPM (Transcripts Per Million) values was constructed by TBtools v2.102 [47]. It was found that 22 *StBBX* genes were expressed in different tissues of potato, but the remaining 8 *StBBX* genes were rarely expressed in potato tissues (Figure 4; Supplementary Table S7). *StBBX14* was highly expressed in leaf. *StBBX09* and *StBBX17* were highly expressed in flower. High expression levels of *StBBX20* were found in root. *StBBX03*, *StBBX04*, *StBBX05*, *StBBX24*, *StBBX06*, *StBBX11*, *StBBX19* and *StBBX31* were highly expressed in stem, leaf, stolon, petiole, stamen and flower, indicating that these genes may play an important role in potato tuberization. *StBBX17* was more highly expressed in tuber sprout, young tuber and stem.

3.7. Expression Analysis of StBBXs Under Cold Stress

Base on the expression pattern of *StBBX* genes under cold stress in our previous study (Supplementary Table S8) [29], six differentially expressed *StBBX* genes were selected to validate by qRT-PCR (Figure 5). The results revealed that the expression levels of *StBBX01*, *StBBX17* and *StBBX23* were significantly increased after 1 h and 6 h of treatment. The expression level of *StBBX07* was highly expressed at 1 h and 8 h. The expression level of *StBBX28* was relatively high at 1 h but then decreased at subsequent cold treatment time points. The expression level of *StBBX05* was significantly increased after 1 h, 6 h and 8 h of treatment.

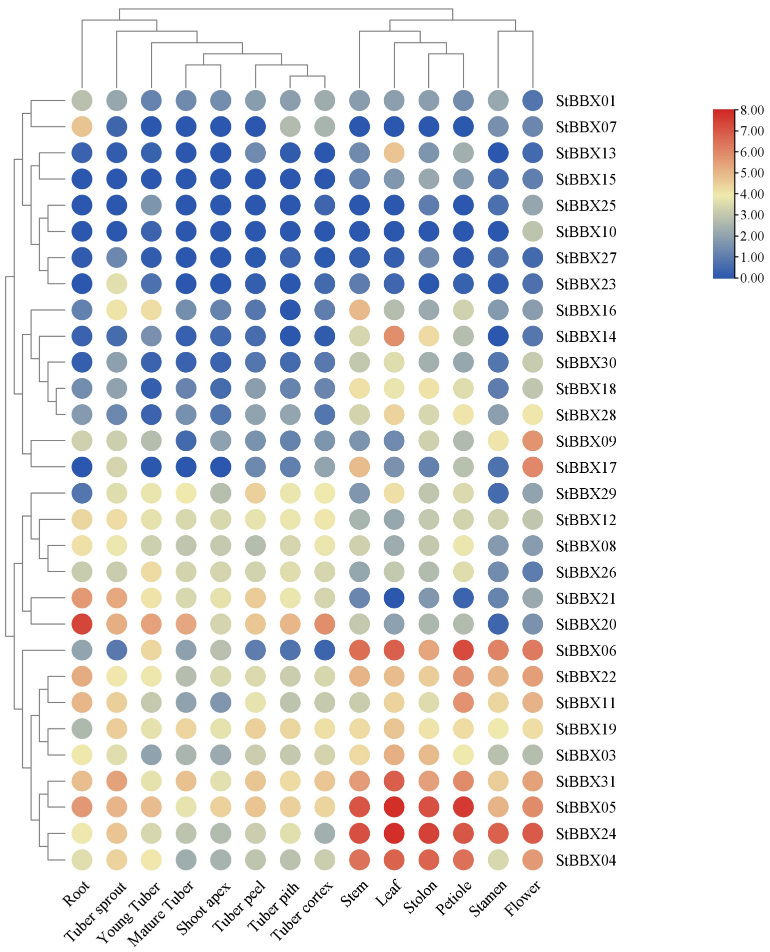

Figure 4. The gene expression levels of *StBBX* genes in different tissues. The expression level of *StBBX* genes in different tissues. The heatmap was constructed based on the TPM values of *StBBX*s.

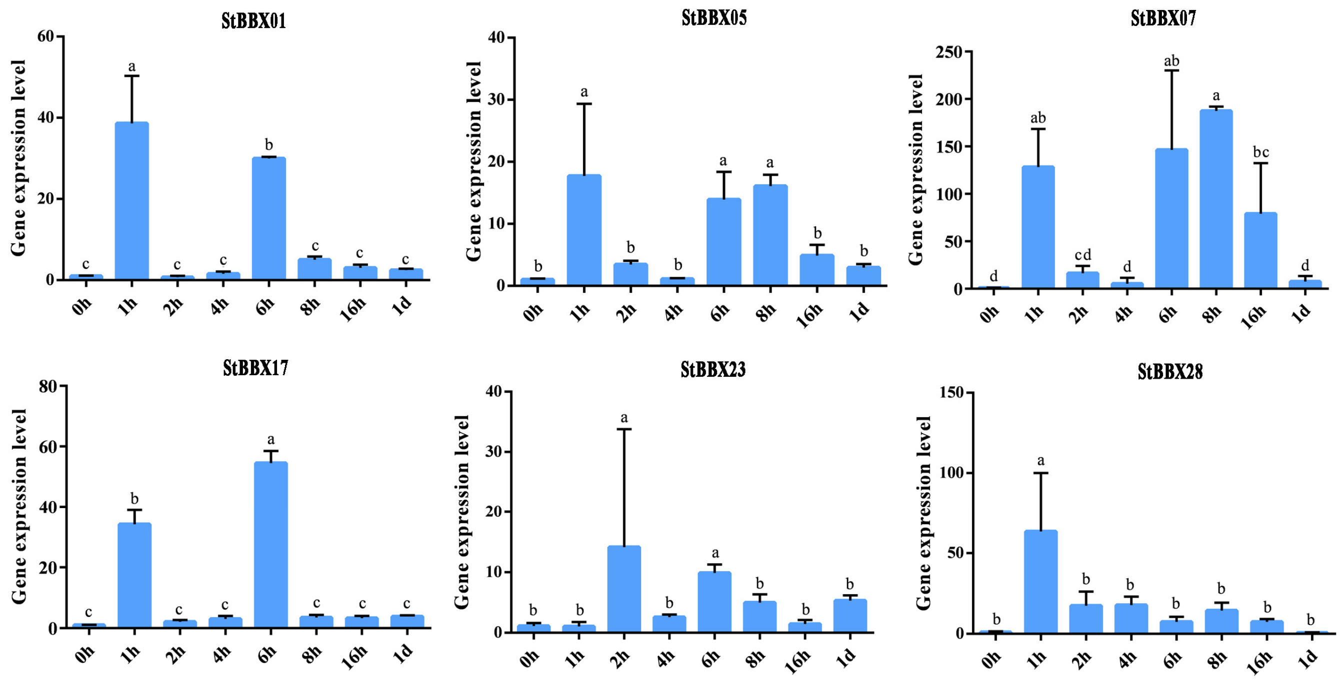

Figure 5. qRT-PCR expression analysis of the StBBX genes in potato Desiree after treatment at 2 °C for 1 h, 2 h, 4 h, 6 h, 8 h, 16 h, and 1 d. The fold change represents units on the Y axis. All data points shown are mean ± SE ($n = 3$). Different lowercase letters indicate a significant difference ($p < 0.05$, one-way ANOVA).

3.8. Subcellular Localization of StBBX17

The coding sequence (CDS) of *StBBX17* contained 393 bp and encoded a protein of 130 amino acids. The CDS of *StBBX17* was cloned in PBWA(V)-GFP constructs and transformed in *Agrobacterium tumefaciens* GV3101. The GV3101 cells contained 35S::PBWA(V)-StBBX17-GFP, and 35S::GFP constructs were injected into *Nicotiana benthamiana*. *StBBX17* did not have an RFP signal on the endoplasmic reticulum. The green GFP signals were observed and overlapped with the nuclear 4′,6-diamidino-2-phenylindole (DAPI) signals. The result indicates that the StBBX17 protein was localized in the nucleus of *N. benthamiana* (Figure 6).

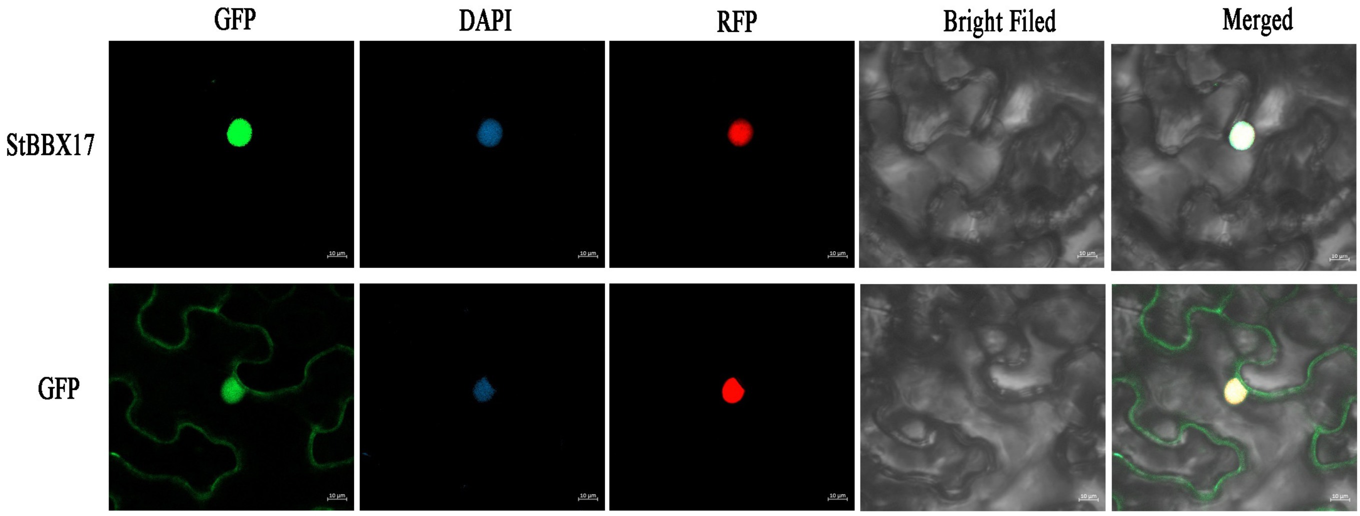

Figure 6. Subcellular localization of StBBX17-GFP and GFP in epidermic cells of *Nicotiana benthamiana* leaves. Bar = 10 μm.

3.9. Overexpression of StBBX17 Enhanced Cold Resistance in Potato

To investigate the function of the *StBBX17* response to cold stress in potato, the overexpression construct of *StBBX17* was transformed into *S. tuberosum* cv. Desiree. A total of 18 transgenic lines of *StBBX17* were obtained by hygromycin resistance. After performing 35S primer validation and measuring gene expression levels, seven positive transgenic lines were obtained. After gene expression analysis of *StBBX17*, two transgenic lines, *StBBX17*-OE-3 and *StBBX17*-OE-23 plants with the highest gene expression levels of *StBBX17*, were selected for further analysis (Figure 7A). The electrolyte leakage of the two transgenic lines was significantly lower than that of wild-type (WT) plants at −1 °C (Figure 7B). Compared with wild-type plants, the two transgenic lines exhibited no morphological changes under normal growth conditions (Figure 7C). The two overexpressing lines and WT plants were subjected to cold treatment (−1 °C for 14 h and recovery for 3 days). The *StBBX17* overexpressing lines showed dramatically improved cold tolerance compared with WT. After cold treatment, the overexpressing lines (*StBBX17*-OE) showed a less injured area of leaves compared with the WT plants (Figure 7D). These results suggest that *StBBX17* acts as a positive regulator of cold tolerance in potato.

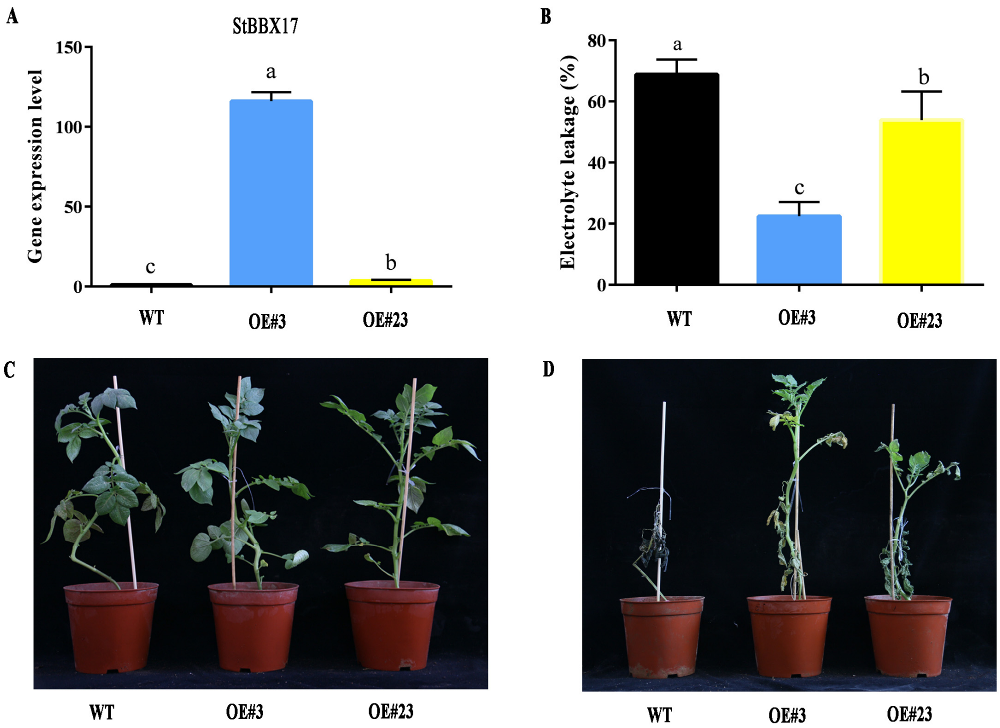

Figure 7. Overexpression of StBBX17 enhanced the cold resistance in potato. (**A**) The gene expression levels of StBBX17 in WT, OE-StBBX17-3 and OE-StBBX17-23 plants. (**B**) The electrolyte leakage of genotypes at the indicated temperatures. (**C**,**D**) The WT and StBBX17 overexpressing transgenic lines exposed to freezing treatment, and after being subjected to −1 °C for 14 h, followed by 3 days of recovery under normal conditions. The different letters represent significant differences at $p < 0.05$ (one-way ANOVA with Tukey's multiple comparisons test).

4. Discussion

The plant gene BBX belongs to the zinc finger gene family and is involved in biological processes such as photoperiod, flowering, cold stress, and anthocyanin [48]. Whole-genome identification of BBX has been widely reported in crops such as *Arabidopsis* [14], rice [49], and tomato [26]. In this study, a systematic whole genome identification and characterization of the BBX gene family were conducted in potato. In the current study, 30 *StBBX* genes were identified in potato; the gene number was consistent with previous studies [28]. Many studies have reported that BBX was mainly classified into five categories [50,51]. In this study, StBBX proteins were grouped into five subfamilies by sequence similarity to Arabidopsis and tomato BBX proteins. The segmental and tandem duplication played an important role in the expansion of the gene family [48,49]. The 15 pairs of *StBBX* genes were identified to duplicated genomic regions, and no tandem duplicated gene was determined, similarly to previous studies [28]. Syntenic analysis among different plant genomes can acquire the conserved features in closely related species [52]. The 52 pairs of orthologous *BBXs* between tomato and potato and 47 pairs of orthologous *BBXs* between *Arabidopsis* and potato were obtained, respectively (Figure 3B). These results provide important information for understanding the functions of *StBBX* genes in potato.

Many studies have revealed that B-box genes play important roles in light-mediated developmental processes and response to cold stress in plants [20,24]. Overexpression of *PpBBX18* promotes anthocyanin accumulation under light treatments [18]. HY5 is a central regulator in light signaling that interacts with BBX21, BBX22, BBX24, BBX25 and BBX28 [53]. The promoters of all *StBBX* genes with light-responsive elements were identified in this study. Compared with wild-type, overexpression of *StCO* restrains potato tuberization under a weakly inductive photoperiod [54]. In this study, the *StCO* gene was *StBBX03*, which was highly expressed in leaf and stolon according to transcriptome data. Compared with WT, silencing of *StBBX24* induced an early flowering phenotype [55]. The *StBBX24* identified in this study was the *StBBX24* identified in previous study. *StBBX09* and *StBBX17* were highly expressed in flower, and *StBBX04*, *StBBX05*, *StBBX06*, *StBBX24*, and *StBBX31* were higher expressed in stem, leaf tuber, stolon, and petiole flower. These *BBX* genes provide data support for further exploration of their functions in potato tuberization. In this study, nine *StBBX* genes contained low-temperature responsive and drought responsive cis-elements, respectively, indicating that they might participate in the response to low-temperature and drought stress.

Tetraploid potato is not tolerant of low temperature and frost, which seriously affects the growth of the above-ground parts of potato and eventually leads to a decrease in yield. Many studies have reported that cold-resistant genes isolated from wild materials were transferred into potato tetraploids, enhancing their cold resistance [34,56]. The genes for cold resistance traits in potato have been reported, but their molecular mechanisms underlying cold stress are not yet fully understood. Previous studies found that the *CRY2-COP1-HY5-BBX7/8* module regulated blue light-dependent cold acclimation in *Arabidopsis* [20]. *StBBX07* was found to homologous to *AtBBX7* and *AtBBX8*, which was differentially expressed under cold stress by qRT-PCR in this study [20]. In apple, *MdBBX37* binds to the *MdCBF1* and *MdCBF4* promoters regulating cold stress tolerance [19]. In this study, no potato StBBX gene homologous to apple *MdBBX37* was found, demonstrating that the BBX gene has a certain specificity among different species. According to the results from the evolutionary tree, *StBBX07*, *StBBX09*, *StBBX20* and *StBBX31* in potato were homologous to *SlBBX7*, *SlBBX9*, *SlBBX20* and *SlBBX31*, which regulated cold tolerance in tomato [24]. The *StBBX29* in potato was homologous to *Arabidopsis* negative regulator *AtBBX29* [25]. qRT-PCR results found that expression of four *StBBX* genes was significantly increased after 1 h; *StBBX01*, *StBBX07*, *StBBX17* and *StBBX23* genes maintained relatively high expression levels at 1 h and 6 h, respectively. *SlBBX17* interacts with *SlHY5* and positively regulates cold tolerance in tomato [24]. *StBBX17* was homologous to *SlBBX17*, and *StBBX17* was differential expressed under cold stress in our previous studies [45]. Compared to baseline at 0 h, *StBBX17* was upregulated in *S. cardiophyllum* under 4 °C treatment for 14 d and exposure to −2 °C for 6 h after cold acclimation for 14 d [45]. In this study, the gene expression levels of *StBBX17* were significantly increased after 2 °C treatment for 1 h and 6 h as compared with 0 h by qRT-PCR results. These results demonstrated that the *StBBX17* gene was upregulated under both diploid and tetraploid cold treatments. Overexpression of *StBBX17* in 'Desiree' corresponded with less leaf damage and relatively low electrolyte leakage as compared with the WT, indicating the *StBBX17* gene positively regulates cold stress in potato. Therefore, the function of the *StBBX* gene under cold stress in potato and whether it depends on the CBF pathway require further study.

5. Conclusions

In this study, 30 StBBX genes were identified in potato. The comprehensive analysis of phylogenetic relationships, conserved domains, chromosome locations, cis-acting regulatory elements and duplication events were performed in the StBBX gene family. The gene expression patterns of all StBBXs were assessed in different tissues using transcriptome data. qRT-PCR analysis indicated that six StBBX genes were differentially expressed under cold stress. Overexpression of StBBX17 enhanced the cold resistance of tetraploid potato. These results provide a theoretical reference for exploring the molecular mechanisms of StBBX genes involved in potato under cold stress and offer theoretical support for further genetic improvement of cold resistance traits in potato.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/horticulturae11101167/s1, Supplementary Figure S1. The motif types in all *StBBX* genes. The *p*-value was used to evaluate the significance of the motif. Supplementary Figure S2. The intraspecific collinearity of the StBBX gene family in the potato genome. Collinear blocks across the entire genome were depicted with a grey background, and red lines represented duplicate StBBX genes. Supplementary Tables S1–S8: Supplementary Table S1. Primer sequences used in this study. Supplementary Table S2. Information and chromosomal location of the StBBX genes. Supplementary Table S3. The infomation of BBXs in Arabidopsis and tomato. Supplementary Table S4. The cis-acting elements detected in the promoter StBBXs genes. Supplementary Table S5. The collinearity orthologous gene pairs among potato, tomato and Arabidopsis. Supplementary Table S6. Whole genome duplication events of StBBX. Supplementary Table S7. The gene expression data of StBBXs in different tissues in potato genome. Supplementary Table S8. The gene expression data of StBBXs in OE-StBBX14 and WT under cold stress (2 °C for 1 and 12 h).

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